Asiful Arefeen

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Summary

– Biomedical Data Science PhD & Computer Science Masters Student at Arizona State University
– Published in ACM Health (I.F. 8.0), CHASE 2022, IEEE BSN 2022, BHI 2023, 2024 (AR 25%), EMBC 2025
– Recipient of **NIH T32 Institutional Training Grant** for AI in Precision Nutrition (AIPrN)
– 5+ years of expertise in conducting ML research in computer vision, timeseries analysis and health data
– Awarded $55,000 in research funding as Champion of 2024 College of Health Solutions Heat and Health Research Challenge
– Experienced in Actionable Counterfactual explanations, explainable AI, human centered AI, multimodal analysis, timeseries forecasting, classification, anomaly detection, transformer models, active learning, transfer learning, finetuning foundation models, multivariate sensor data, model deployment on micro-controller, GenAI
– Collaborated with interdisciplinary teams: endocrinologists, nutritionists, data scientists, front-end engineers.
– **As a US citizen, I do not require visa sponsorship**

Education

Arizona State University — Aug 2021 - May 2026

PhD in Biomedical Informatics & Data Science — *Aug 2021 - May 2026*
MS in Computer Science — *Jan 2024 - Dec 2025*
MS in Biomedical Informatics — *Aug 2021 - May 2023*

- **Advisor:** Dr. Hassan Ghasemzadeh – website: ghasemzadeh.com
- **PhD Thesis:** Toward AI-Driven Interventions for Improved Health Outcomes
- **MS Thesis:** Label-Efficient Artificial Intelligence in Digital Health
- **Notable Courses:** Data Structures and Algorithms, Machine Learning, Neural Network Design, Embedded Machine Learning, Data Mining, Linear Algebra, Knowledge Representation & Reasoning

Bangladesh University of Engineering & Technology — Feb 2015 - Apr 2019

BS in Electrical & Electronic Engineering

- **Thesis:** Measuring and assessing the effects of harmonics on the power grid during EV charging
- Published three papers in IEEE conference proceedings

Research Interest

- Explainable AI (XAI) for healthcare: designing AI models that are trustworthy and provide transparent and interpretable treatment recommendations to clinicians.
- Counterfactual treatment recommendations: generating “what-if” scenarios to explore alternative interventions for patient care.
- Generative AI for healthcare: creating synthetic patient data, clinical simulations, or predictive scenarios to support counterfactual decision-making.
- AI-driven intervention design: optimizing personalized treatment plans based on patient-specific data and outcomes.
- Digital twins for treatment simulation: building virtual patient models to simulate and predict responses to therapies.
- Label-efficient AI in digital health: using active learning, self-supervised learning to develop models that perform well with limited labeled clinical data.
- Mobile health (mHealth) applications: leveraging smartphones, wearable sensors and apps for remote monitoring, patient engagement and passive sensing for chronic disease management.
- Computer vision for diagnostics: analyzing medical images (X-ray, MRI, CT, Ocular) to support disease detection and diagnosis.
- Large Language Model (LLM) fine-tuning: adapting pre-trained language models to extract and summarize healthcare knowledge from clinical notes.

- Data exploitation and visualization: transforming complex healthcare datasets into actionable insights for clinicians.
- OCR for unstructured clinical data: extracting structured information from handwritten or scanned medical records.

Experience

Mayo Clinic, Arizona | *Research Affiliate* Oct 2023 – Present
– Developed *GlyMan*, a counterfactual intervention technique to assist current automated insulin delivery (AID) technology in diabetes management. GlyMan surpasses baselines (DiCE, NICE, CFNOW) in validity and proximity. Working on GlyTrace, an end-to-end counterfactual explanations software to assist physicians in clinical decision making.

Embedded Machine Intelligence Lab, ASU | *Grad Research Assistant* Aug 2021 – Present
– Developed novel counterfactual technique *ExAct* for behavioral intervention towards better health outcome. Performance wise, ExAct improves validity by at least 10% and proximity by at least 6.6% compared to baselines like DICE, NICE, CEML, and Optbinning.

Washington State University | *Grad Teaching & Research Assistant* Aug 2020 – July 2021
– Performed research on motion artifact suppression in physiological signals. TA for Computer Security, Data Structures C/C++, Program Design and Development. Held lab sessions and office hours. Set and graded quizzes, graded assignments.

Awards & Achievements

Fellowship: NIH T32 Institutional Training Grant for AI in Precision Nutrition (AIPrN) Research (2024-ongoing)

Research Grant: Champion of 2024 College of Health Solutions (CHS) Student Heat and Health Research Challenge Award. Awarded $55,000 in research funding.

Best Poster: Best poster award at the 2024 College of Health Solutions (CHS) Faculty Research Day.

Travel Award: NSF Student Travel Award to attend IEEE BHI 2024, BHI 2023, ACM CHASE 2022, ASU GSG Travel Award for EMBC 2025 in Copenhagen, Denmark, 2x ASU CHS Travel Support in 2025 for EMBC 2025 and BSN 2025.

Grant: ASU Graduate College University Grant 2022-23, 2023-24

Teaching Experience

1. **Instructor,** BMI 310: App Development for Population Health, Arizona State university, Spring 2024.
2. **Guest Lecturer,** Wearable Sensing and Machine Learning for Precision Nutrition, BMI 570: BMI Seminar Series, Arizona State university, Spring 2023.

Invited Talks

1. Towards Generating User-Centric and Realistic Counterfactuals for Positive Health Outcomes, Truveta, Seattle, WA, July 2025.
2. Towards AI-Driven Behavioral Interventions in Digital Twin Systems for Improved Health Outcomes, Control Systems Engineering Laboratory (CSEL), Tempe, AZ, June 2025.

Publications

Peer reviewed Journals

1 Mamun, A., **Arefeen, A.**, Racette, S.B., Sears, D.D., Whisner, C.M., Buman, M.P., & Ghasemzadeh, H. (2025). LLM-Powered Prediction of Hyperglycemia and Discovery of Behavioral Treatment Pathways from Wearables and Diet. *Sensors (Basel, Switzerland)*, 25.

2 **Arefeen, A.**, & Ghasemzadeh, H. (2025). Cost-Effective Multitask Active Learning in Wearable Sensor Systems. *Sensors (Basel, Switzerland)*, 25.

3 **Arefeen, A.**, Akbari, A., Mirzadeh, S., Jafari, R., Shirazi, B., & Ghasemzadeh, H. (2023). Inter-Beat Interval Estimation with Tiramisu Model: A Novel Approach with Reduced Error. *ACM Transactions on Computing for Healthcare.*

4 Mirzadeh, S., **Arefeen, A.**, Ardo, J., Fallahzadeh, R., Minor, B.D., Lee, J., Hildebrand, J.A., Cook, D., Ghasemzadeh, H., & Evangelista, L.S. (2022). Use of machine learning to predict medication adherence in individuals at risk for atherosclerotic cardiovascular disease. *Smart health, 26*.

Peer reviewed Conferences

5 Soumma, S.B., **Arefeen, A.**, Carpenter, S.M., Hingle, M., & Ghasemzadeh, H. (2025). SenseCF: LLM-Prompted Counterfactuals for Intervention and Sensor Data Augmentation. In Proceedings of the *IEEE-EMBS International Conference on Body Sensor Networks (BSN'25).*

6 Khamesian, S., **Arefeen, A.**, Thompson, B., Grando, M.A., & Ghasemzadeh, H. (2025). AZT1D: A Real-World Dataset for Type 1 Diabetes. In Proceedings of the *IEEE-EMBS International Conference on Body Sensor Networks (BSN'25).*

7 **Arefeen, A.**, Fessler, S.N., Mostafavi, S.M., Johnston, C., & Ghasemzadeh, H. (2025). MealMeter: Using Multimodal Sensing and Machine Learning for Automatically Estimating Nutrition Intake. In Proceedings of the *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25).*

8 **Arefeen, A.**, & Ghasemzadeh, H. (2025). LEAD: Localized explanations with adversarial decision boundary characterization for interpretable disease prediction. In Proceedings of the *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25).*

9 **Arefeen, A.**, Khamesian, S., Grando, M.A., Thompson, B., & Ghasemzadeh, H. (2024). GlyMan: Glycemic Management using Patient-Centric Counterfactuals. *2024 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 1-5.

10 Shroff, P., **Arefeen, A.**, & Ghasemzadeh, H. (2023). GlucoseAssist: Personalized Blood Glucose Level Predictions and Early Dysglycemia Detection. *2023 IEEE 19th International Conference on Body Sensor Networks (BSN)*, 1-4.

11 **Arefeen, A.**, & Ghasemzadeh, H. (2023). GlySim: Modeling and simulating glycemic response for behavioral lifestyle interventions. *2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 1-5.

12 **Arefeen, A.**, Fessler, S.N., Johnston, C., & Ghasemzadeh, H. (2022). Forewarning Postprandial Hyperglycemia with Interpretations using Machine Learning. *2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, 1-4.

13 **Arefeen, A.**, Jaribi, N., Mortazavi, B.J., & Ghasemzadeh, H. (2022). Computational Framework for Sequential Diet Recommendation: Integrating Linear Optimization and Clinical Domain Knowledge. *2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 91-98.

14 Alinia, P., Parvaneh, S., Mirzadeh, S., **Arefeen, A.**, & Ghasemzadeh, H. (2022). Boosting Lying Posture Classification with Transfer Learning. *44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'22)*, 109-114.

15 **Arefeen, A.**, Shehreen, S., Islam, A.K., Rahman, M., & Rahman, H. (2019). Measuring and Assessing the Effects of Harmonics on the Grid of Bangladesh Power System During EV Charging. *2019 IEEE International Conference on Power, Electrical, and Electronics and Industrial Applications (PEEIACON)*, 5-9.

Preprints

16 **Arefeen, A.**, Soumma, S.B., & Ghasemzadeh, H. (2025). RealAC: A Domain-Agnostic Framework for Realistic and Actionable Counterfactual Explanations. *ArXiv, arXiv:2508.10455.*

17 **Arefeen, A.**, Khamesian, S., Grando, M.A., Thompson, B., & Ghasemzadeh, H. (2025). GlyTwin: Digital Twin for Glucose Control in Type 1 Diabetes Through Optimal Behavioral Modifications Using Patient-Centric Counterfactuals. *ArXiv, abs/2504.09846*.

18 Khamesian, S., **Arefeen, A.**, Grando, M.A., Thompson, B., & Ghasemzadeh, H. (2025). Type 1 Diabetes Management using GLIMMER: Glucose Level Indicator Model with Modified Error Rate. *ArXiv, abs/2502.14183*.

19 **Arefeen, A.**, & Ghasemzadeh, H. (2023). Designing User-Centric Behavioral Interventions to Prevent Dysglycemia with Novel Counterfactual Explanations. *ArXiv, abs/2310.01684*.

Conference Presentations

Oral

1 LEAD: Localized Explanations with Adversarial Decision Boundary Characterization for Interpretable Disease Prediction at the 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25), Copenhagen, Denmark.

2 GlySim: Modeling and Simulating Glycemic Response for Behavioral Lifestyle Interventions at the 2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'23), Pittsburgh, PA, USA.

3 Computational Framework for Sequential Diet Recommendation: Integrating Linear Optimization and Clinical Domain Knowledge at the 2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE'22), Washington, DC, USA.

Poster

4 MealMeter: Using Multimodal Sensing and Machine Learning for Automatically Estimating Nutrition Intake. at the 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC'25), Copenhagen, Denmark.

5 GlyMan: Glycemic Management using Patient-Centric Counterfactuals at the 2024 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI'24), Houston, TX, USA.

Demo

6 GlucoGuide: A Retrospective Counterfactual Decision Support to Address Stakeholder Needs at the 2025 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN'25), Los Angeles, CA, USA.

Professional Services

- Journal reviewer for ACM Transactions on Computing for Healthcare (HEALTH).
- Journal reviewer for IEEE Journal Of Biomedical & Health Informatics (JBHI)
- Journal reviewer for ACM Transactions on Intelligent Systems and Technology
- Reviewer for International Conferences of the IEEE Engineering in Medicine and Biology Society (EMBC 2025)
- Reviewer for International Conferences of the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2023, 2024, 2025)
- Reviewer for International Conferences of the IEEE International Conference on Pervasive Computing and Communications (PerCom 2023)

Mentorship Experience

Research Mentor, Arizona State University: One Master's student and one High-School student

1. Aashritha Machiraju (MS, CS, 2024-Present), in thesis formulation and conference publication.
2. Prisha Shroff (Hamilton High School, Chandler, AZ, 2023, current affiliation: undergrad student in Computer Science at Stanford), mentorship led to publishing a research paper on glycemic management using AI (GlucoAssist in IEEE BSN 2023).

Ongoing Projects

RealAc: Towards Realistic and User-Centric Counterfactual Explanations| *GitHub*

- RealExAct aims to generate counterfactual explanations that can automatically sense and preserve the non-linear inter-feature dependencies exist in the real data. Additionally, RealExAct preserves user preferences to suppress the change of certain feature groups and make the counterfactuals actionable.

GlyTwin: Digital Twin for Glucose Control in Type 1 Diabetes Through Behavior Change | *GitHub*

- We enhance the capabilities of traditional digital twin paradigm with counterfactual interventions/explanations. We are working on getting it validated with physicians and users.

FoodOCR: Information Extraction from Handwritten Food Logs | *GitHub*

- In this project, we had several participants who logged their meals on paper. Later, we used Google Cloud Vision OCR to extract the handwritten food logs. As the extractions were not 100% accurate, the extracted texts were fixed using BERT through prompt engineering.

GlySynth: Autoregressive Framework for Synthetic Glycemic Response Generation | *GitHub*

- A conditional autoregressive scheme for generating synthetic physiological signals (preferably glycemic response) given contextual information. As validated by downstream classification task, GlySynth produces better signals compared to GANs under data scarcity.